QUALITY INFORMATION DOCUMENT

For the Ocean Monitoring Indicator on the Black Sea Overturning Circulation

Issue: 1.0

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RECORD TABLE

When the quality of the products changes, the QuID is updated and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

Issue	Date	§	Description of Change	Author	Validated By
1.0	2024/11/26	All	Initial version	L. Lima, M. Ilicak, E. Jansen	

TABLE OF CONTENTS

RECORD TABLE **2**

1) INTRODUCTION **4**

a) Product Name 4

b) Indicator Evaluation Method 4

c) Copernicus Marine products used as input for the computation 5

2) SCIENTIFIC VALIDATION **6**

a) Black Sea Overturning Circulation Indicator 6

3) REFERENCES **7**

1) INTRODUCTION

a) Product Name

OMI_CIRCULATION_MOC_BLKSEA_area_averaged_mean.

b) Indicator Evaluation Method

To understand the overturning circulation of the Black Sea, we compute the residual meridional overturning circulation in density space. Residual overturning as a function of latitude (y) and density ($\bar{\sigma}$) bins can be computed as follows:

$$\psi^*(y,\bar{\sigma}) = -\frac{1}{T}\int_{t_0}^{t_1}\int_{x_{B1}}^{x_{B2}}\int_{-H}^{0}\mathcal{H}[\bar{\sigma}-\sigma(x,y,z,t)]\times v(x,y,z,t)dzdxdt,$$

where $\mathcal{H}$ is the Heaviside function and v is the meridional velocity. We used 100 σ_2 (potential density anomaly with reference pressure of 2000 dbar) density bins to remap the mass flux fields.

c) Copernicus Marine products used as input for the computation

Product name (and Type)	Documentation
BLKSEA_MULTIYEAR_PHY_007_004	QUID: https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-BLK-QUID-007-004.pdf PUM: https://catalogue.marine.copernicus.eu/documents/PUM/CMEMS-BLK-PUM-007-004.pdf

2) SCIENTIFIC VALIDATION

a) Black Sea Overturning Circulation Indicator

More details and full scientific evaluation can be found in the Copernicus Marine Ocean State Report issue 6 (OSR6, von Schuckmann et al., 2022) Section 2.9 (Ilicak et al., 2022). With respect to the results shown in OSR6, an updated version of the Black Sea Reanalysis has been used (version E4R1) to evaluate the present ocean monitoring indicator (OMI). This version has a different spinup compared to the OSR6 (version E3R1).
§
The cold intermediate layer (CIL) and the inflow of Mediterranean Water (MW) through the Bosphorus Strait are critical processes that ventilate the subsurface layers in the Black Sea (Özsoy et al., 1993; Özsoy and Ünlüata, 1997). Recent evidence suggests that human-induced climate change is driving environmental changes in the Black Sea (Stanev et al., 2019; Capet et al., 2020). Notably, observations have revealed a significant decline in the CIL over the past two decades. However, understanding the impact of changes in water mass properties, such as the CIL and MW, on the circulation patterns of the Black Sea remains a complex and ongoing challenge.

The Black Sea Meridional Overturning Circulation (BS-MOC) OMI is essential for understanding the evolving dynamics of intermediate water masses in the Black Sea. The development of a robust index could be instrumental in monitoring long-term indicators of the Black Sea's ecosystem health, particularly in relation to anthropogenic climate change. The main reason undertaking the present OMI is to develop an index that is possible to assess a future climate change on the Black Sea circulation. The overturning index that gives information about surface, sub-surface and deep ocean circulation is unique. Therefore, this new index will trace possible buoyancy forcing changes in terms of the Bosphorus Strait inflow, river runoff into the Black Sea and sea surface temperature increase trend, and their impact onto the circulation.

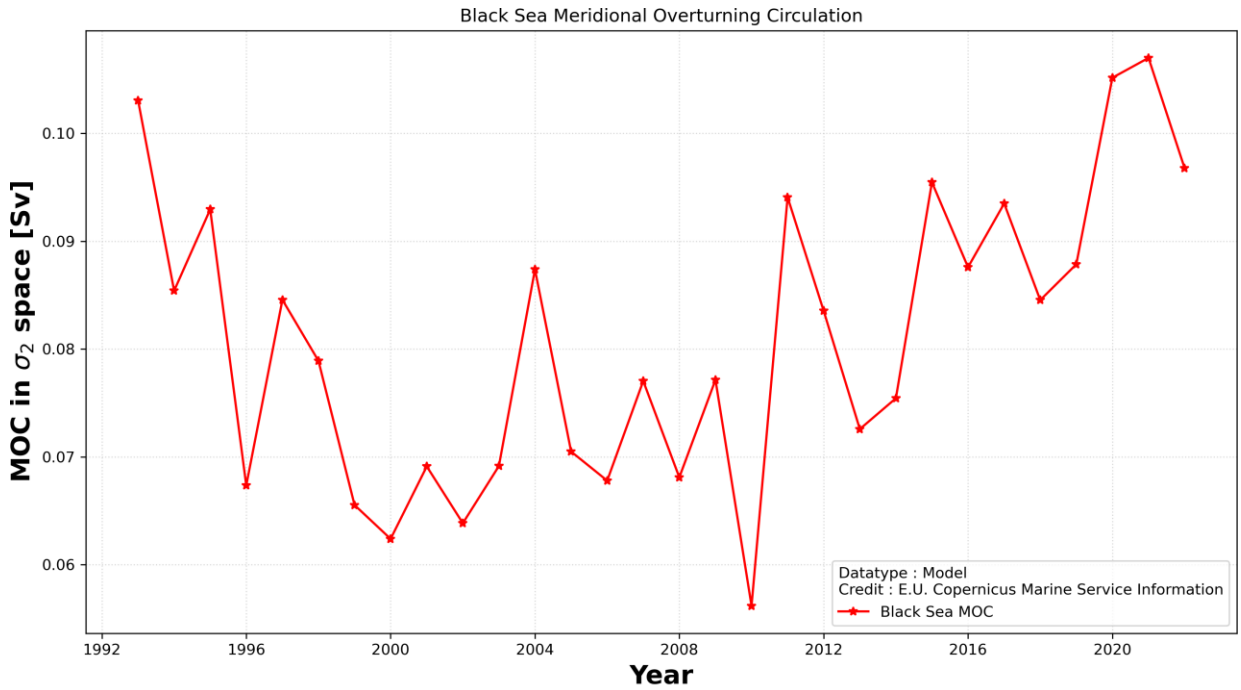

Figure 1: *Time evolution of the maximum BS-MOC in density space between 22.45 and 23.85 kg m^{-3}.*

3) REFERENCES

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